

# Design and Dynamic Analysis of Random Vibration Test Fixture for Aerospace Valve

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**Abstract**—The aerospace valve is an important part of the rocket system, and its working state is greatly affected by random vibration. In the ground environment vibration test, how to design a high performance fixture to make the valve and the vibration table connected closely and transmit the complete vibration load is very important. This paper uses the combination of T-shaped fixture and threaded connection to design a fixture that can undertake valve environmental vibration test, and conduct dynamic analysis. Through modal analysis, the natural frequency and vibration mode of the fixture system are obtained, and the key position is obtained through random response analysis. The output response is judged whether it meets the requirements of transmitting a complete vibration load. Because the fixture bolt layout has a great impact on the performance, three groups of dynamic analysis were carried out by changing the bolt spacing to compare the results, and the impact of the bolt layout on the fixture was analyzed.

**Keywords**- modal analysis; Random response; fixture design; Bolt layout

## I. INTRODUCTION

Aerospace valve is an important component in the rocket system and is a typical mechanical structural part. Its reliability directly affects the success of the rocket launch mission, and even directly relates to the safety of personnel and equipment. Therefore, the system proposes a very high reliability index for the valve.

The low-frequency vibration felt by astronauts in China's first manned space flight is the self-excited vibration generated by the coupling of the propulsion system and the vibration of the arrow body structure, and these vibrations and shocks will also bring very serious harm to the rocket structure, even destroy the entire rocket, and eventually lead to the failure of the launch mission [1]. According to NASA's investigation report on the causes of satellite launch failures, 45% of satellite launch

failures are caused by structural vibrations or sound-induced vibrations during the launch phase of the rocket [2]. In addition, the rocket's vibration characteristic test shows that the payload and engine system have their own vibration patterns in the full rocket vibration, especially the two-stage rocket [3].

In order to solve these problems, people have introduced the principles of statistical mechanics and noise theory, and the subject of random vibration has gradually formed. Due to the limitation of test site, time, funds, most environmental vibration tests adopt laboratory simulation, which is an important research method to assess and evaluate the adaptability of products to the vibration environment. Through the vibration table to simulate the random excitation caused by the rocket engine, the valve is subjected to random vibration test to determine its reliability index.

Combined with the working environment characteristics of a space electric air valve and the influencing factors considered in reliability analysis and life assessment, and considering vibration, temperature, working load, a comprehensive environmental performance degradation and reliability test platform for the electric air valve is built, as shown in the Fig. 1 below.

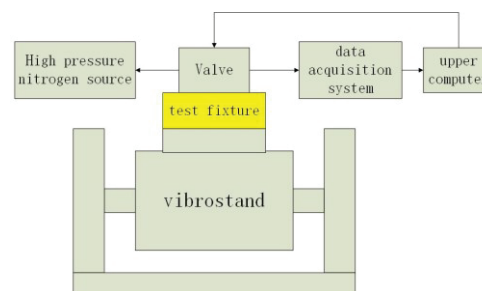


Figure 1. Overall structural principle of the test system

The implementation process of environmental vibration test usually involves the induction and formulation of test conditions, vibration fixture design and simulation analysis, sensor layout, etc. At the same time, there are also some problems that cannot be described in detail in standards and regulations during the test implementation, such as the fixture cannot satisfy the stable fixation with the shaking table, and the non-uniformity of vibration transfer seriously affects the credibility of the test results. Therefore, the dynamic analysis of the fixture system based on the finite element software helps us get whether the response of the fixture system meets the test requirements, and it can also improve the fixture structure to make the result more ideal. Regarding the design of the vibration test fixture, Liu Kai analyzed the natural frequency and vibration mode of a vibration fixture for a car lamp based on the ABAQUS software, and performed a structural optimization design of the fixture [4]. Hu Yongxiang modal analysis of a model of the three-axis turntable is carried out, it is concluded that the resonance frequency and vibration mode of turntable, and carry on the improvement of the structural [5]. For the vibration test of a large slender eccentric cabin, Hang Chao designed its vibration test fixture using the counterweight method. And through the finite element software ABAQUS, the modal parameters and random vibration response of the fixture in the free state and the connected test piece state are analyzed [6].

Perform dynamic analysis on the fixture and valve connection system. First of all, modal analysis is used to extract the modal parameters of the structure-natural frequency and mode shape. Its purpose is to avoid resonance and harmful mode shapes when the system is working. Judge whether the structure design is reasonable and whether there are any weak links in the structure. It can also be used as the starting point for other more detailed and in-depth dynamic analysis. Secondly, the random response analysis is mainly to analyze whether the dynamic response of the clamp and the valve connection position, such as acceleration response, can meet the excitation transmission requirements under the action of external excitation load.

This fixture is designed based on simple manufacturing and low cost, using T-shaped and bolted connections, and using CATIA software to establish fixtures and certain three-dimensional solid model of electric valve structure. Using the theory of dynamic modal analysis, combined with the Lanczos calculation algorithm in the finite element software ABAQUS, the dynamic characteristics of the structure are analyzed and researched, and the modal parameters of the structure-natural frequency and mode shape are extracted. Based on the random vibration theory, the random response analysis of the structure is carried out, and the acceleration response curve and the acceleration root mean square response curve of the key position points are obtained. For bolted fixtures, the bolt layout has a greater impact on the overall dynamic performance, so by changing the bolt layout many times, dynamic simulation analysis is carried out, and the output response results can not only improve the fixture, but also analyze the impact of the bolt layout on the fixture.

## II. FIXTURE DESIGN

The fixture in the vibration test is one of the key components for the test piece to perform the vibration test smoothly. In the

test, the fixation of the test piece and the transmission of the vibration load are all done by the fixture [4], and the pros and cons of its dynamic characteristics (this article refers to the amplitude-frequency characteristics of the vibration excitation response) are directly related to the authenticity of the test results. In addition to the ideal vibration test fixture to meet the boundary constraint conditions, the transfer of force, the combination of the center of gravity of the specimen and the moving coil, the strength of the fixture, stiffness and test equipment capacity limitations and other requirements, but also requires that the vibration excitation is transmitted to the specimen without distortion, that is, the load imposed by the fixture to the specimen should be consistent with the set reference spectrum.

### A. Base connection design

For the vibration table system, the vibration table moving coil and table surface have high rigidity and high natural frequency. The fixture base and the vibration table surface are connected by multiple bolts, and the connecting screw holes are usually in the form of counterbores. Due to the test requirements, the position of the hole meets the connection of the two types of test shaker, and the fixture installation holes provided by the shaker should be used as much as possible to improve the rigidity of the connection between the fixture and the shaker.

### B. Structural design

Generally, the overall casting process is the first choice, followed by welding and threaded connection technology. However, the casting process takes a long time and the cost is high. Welding tends to deal with the deformation of the connection part. Therefore, this clamp adopts a combination of T-shaped and threaded connection. This clamp can ensure the symmetry of the structure and good vibration stability, which can make the clamp and the combined center of gravity of the test piece is located on the central axis of the table, thereby reducing the impact of the test process due to the center of gravity does not overlap the impact of shaking. It can also ensure convenient processing and good manufacturability, avoid the phenomenon that welding makes the structure of the connection position uneven, and at the same time make the price much lower than the casting process, and the processing cycle is short. The bottom plate is directly connected to the shaking table, the vertical center plate is designed above the fixture to fix the valve sample, and meet the two working conditions of the electric air valve, and the rest of the parts are fixed by bolt connection. Since the bolt layout has a greater impact on the overall energy transfer of the fixture, the best design plan was found by optimizing the bolt layout in the later stage.

### C. Analysis of quality influencing factors

In principle, the weight of the fixture should be light, because the weight of the fixture directly affects the total weight of the mobile system, and affects the maximum acceleration value that the vibration table can reach after the test piece is installed. The greater the weight of the fixture, the lower the output vibration acceleration of the vibration table. In order to maximize the function of the vibration table, the weight of the fixture should be as low as possible while ensuring the rigidity. The fixture material should be selected with high rigidity, high damping, and

light weight. The material of this fixture is mainly aluminum alloy. The formula analysis is as follows:

$$F = (M_1 + M_2 + M_3) \times a \quad (1)$$

In (1):  $F$  is the rated thrust of the shaker;  $M_1$  is the mass of the moving coil;  $M_2$  is the mass of the test piece;  $M_3$  is the mass of the fixture;  $a$  is the acceleration. Under the same excitation input conditions, the deformation of the test piece is inversely proportional to the stiffness of the test piece [5], the specific formula is as follows:

$$k = \frac{p}{\delta} \quad (2)$$

In (2),  $k$  is the stiffness of the test piece;  $p$  is the force acting on the structure;  $\delta$  is the deformation due to the force. Strain is the relative amount of local deformation of an object under the action of external force and non-uniform temperature field. Displacement is the absolute amount of local deformation of an object under the action of external force and non-uniform temperature field. From (2), it can be obtained that to improve the stiffness of the test piece, it is necessary to reduce the strain and displacement variation. According to the strength theory  $f = \sqrt{k/m}$ , the natural frequency  $f$  is proportional to the stiffness  $k$  and inversely proportional to the mass  $m$  [4]. Therefore, to improve the rigidity, it is necessary to increase the natural frequency of the test piece.

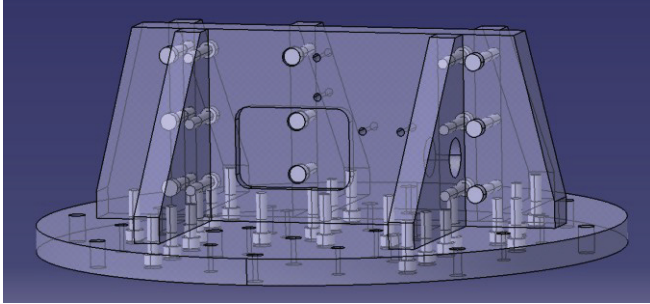


Figure 2. Fixture diagram for environmental vibration test of valve

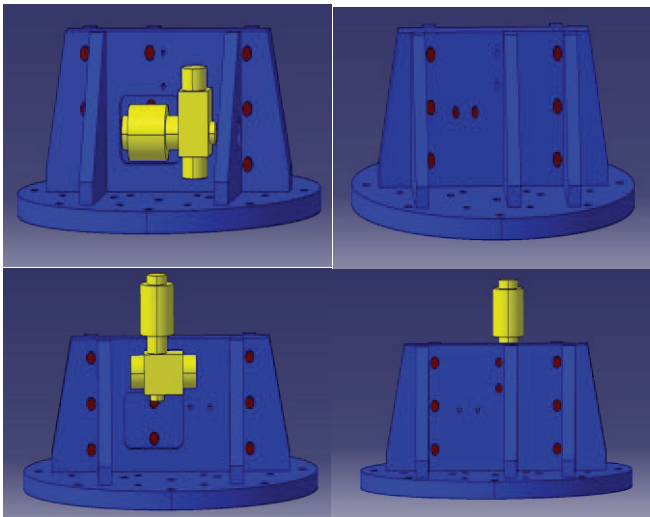


Figure 3. Working status of fixtures and valves

### III. MODAL ANALYSIS OF SYSTEM STRUCTURE

Starting from the mathematical concept, the structural mode is a characterization of the vibration characteristics, which is determined by the eigenvalues and eigenvectors of the system. Therefore, to solve the natural frequency and mode shape of the system is to solve the eigenvalues and eigenvectors of the system. The vibration behavior of the fixture is a typical dynamic problem, and the balance equation it follows is:

$$M\ddot{X}(t) + C\dot{X}(t) + KX(t) = F(t) \quad (3)$$

In (3),  $M$ 、 $C$ 、 $K$  are the system mass matrix, damping matrix and stiffness matrix respectively, and  $\ddot{X}(t)$ 、 $\dot{X}(t)$ 、 $X(t)$  are the system acceleration vector, velocity vector and displacement vector respectively  $F(t)$  is the external force vector of the system. For both ends of the above formula to carry out the transformation, there are:

$$(K - \omega^2 M + j\omega C)X(\omega) = F(\omega) \quad (4)$$

In (4),  $\omega$  is the characteristic value of this equation. Then, the fixed frequency of the original structure can be solved,  $f_i = \frac{\omega_i}{2\pi}$  and the corresponding vibration mode  $\varphi$ .

#### A. (1) Establishing the finite element model

In order to reduce the operating difficulty of the ABAQUS software, the factors that have little influence on the finite element analysis results in the structure are ignored. For example, the threads and chamfers of bolts in valves and fixtures can be ignored. Considering the subsequent mesh division, the top of the hex socket bolt can be unified into a cylinder. The valve structure is simplified, and the valve quality and the position of the center of gravity remain unchanged by changing the density. Import it into the ABAQUS finite element analysis software, and the material property settings are shown in the Table I :

TABLE I. MATERIAL PROPERTIES OF PARTS

| Name    | Materials       | Density (t/mm <sup>3</sup> ) | Young modulus (MPa) | Poisson's ratio |
|---------|-----------------|------------------------------|---------------------|-----------------|
| fixture | Aluminum alloy1 | $2.7 \times 10^{-9}$         | 70000               | 0.3             |
| valve   | Aluminum alloy2 | $3.5 \times 10^{-9}$         | 70000               | 0.3             |
| bolt    | Steel           | $7.9 \times 10^{-9}$         | 206000              | 0.3             |

When interaction is applied to each component, different plates are set to face-to-face contact, the tangential behavior is set to a friction coefficient of 0.3, and the normal behavior is set to hard contact. The base and the middle vertical plate are defined as the main surface, and the side plate is defined as the slave surface. Set the binding constraints of the bolts and the two plates respectively, and set friction with the bottom plate. Define the boundary conditions and set the fixture base to be fully fixed. When meshing, use modified quadratic tetrahedral elements.

#### B. Modal analysis results

In general finite element analysis, the system has many degrees of freedom. At the same time, when studying the system response, it is often necessary to understand only a few lower-



order eigenvalues and corresponding eigenvectors. In the meta-analysis, some more efficient solutions have been developed to adapt to the above characteristics. In this paper, the Lanczos method is used to solve the inherent dynamic characteristics of the fixture structure, and the first 20 natural frequencies and natural modes of the system structure are obtained by calculation, as shown in the Table II and Fig. 4-6.

TABLE II. THE FIRST 20 NATURAL FREQUENCIES OF THE OVERALL SYSTEM STRUCTURE (UNIT Hz)

|           |        |        |        |        |        |
|-----------|--------|--------|--------|--------|--------|
| Degree    | 1      | 2      | 3      | 4      | 5      |
| Frequency | 1285.8 | 1381.7 | 2853.2 | 3210.6 | 3510.1 |
| Degree    | 6      | 7      | 8      | 9      | 10     |
| Frequency | 3604.5 | 3802.4 | 3925.8 | 4038.0 | 4087.5 |
| Degree    | 11     | 12     | 13     | 14     | 15     |
| Frequency | 4604.6 | 4822.0 | 5224.4 | 5500.7 | 5764.0 |
| Degree    | 16     | 17     | 18     | 19     | 20     |
| Frequency | 6827.4 | 7207.8 | 7326.4 | 7741.0 | 8208.0 |

From the results, we can see that the first-order and second-order natural frequencies are in the first 2000 Hz. These two frequency values are relatively close because the deformation of the vibration mode structure is almost the same, and with the increase of the vibration frequency, its own structure changes significantly and the mass distribution becomes uneven, which is also the reason for the large variation of the subsequent natural frequencies.

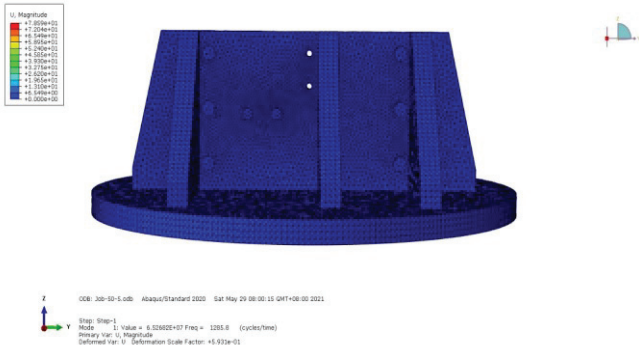


Figure 4. The first-order mode shape



Figure 5. The second-order mode shape

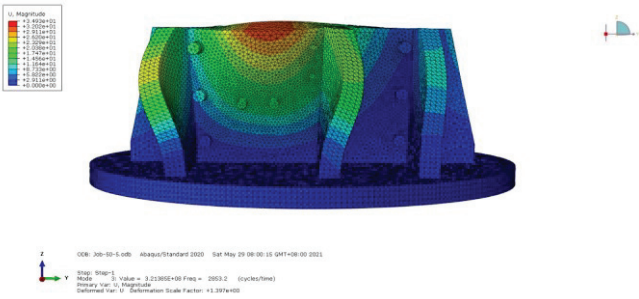


Figure 6. The third-order mode shape

#### IV. RANDOM RESPONSE ANALYSIS OF SYSTEM STRUCTURE

The most obvious feature of random process is that it is non-periodic and its instantaneous value cannot be predicted, so it needs to be carried out by statistical methods. At present, the more widely used indicators used to describe the random response process are: root mean square value, power spectral density [6-8]. In view of the complexity of the stochastic process, it is usually assumed to be a stationary and normal distribution. In the analysis method of power spectral density (PSD), the power spectral density function is an important parameter describing the elapsed process of each state in the steady state, and it directly describes the energy distribution on each frequency component [9]. The root mean square value represents the concentration of random variables near the average value, see (5):

$$\sqrt{\bar{x}^2} = \sqrt{\frac{1}{T} \int_0^T x^2(t) dt} \quad (5)$$

And the probability density of the amplitude indicates that the instantaneous amplitude of random vibration falls within a certain interval Probability, see (6):

$$PSD = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \frac{x^2(t)}{\Delta f} dt \quad (6)$$

##### A. Parameter calculation

In the vibration test, the input is the acceleration power spectral density signal. The slope of the first rising spectrum is 3db/oct, and the slope of the falling spectrum at the end is -6db/oct. Through calculation, the initial coordinate is (20,0.04), and the final coordinate is (2000,0.05).

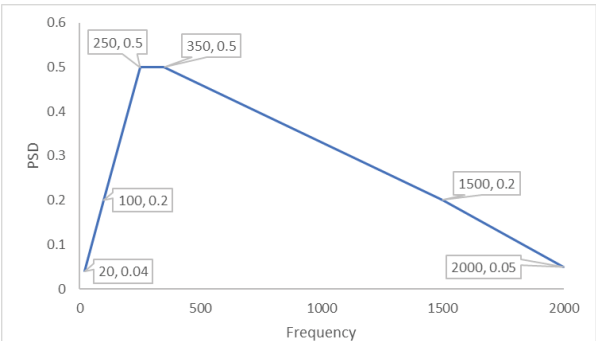


Figure 7. Vibration power spectral density curve

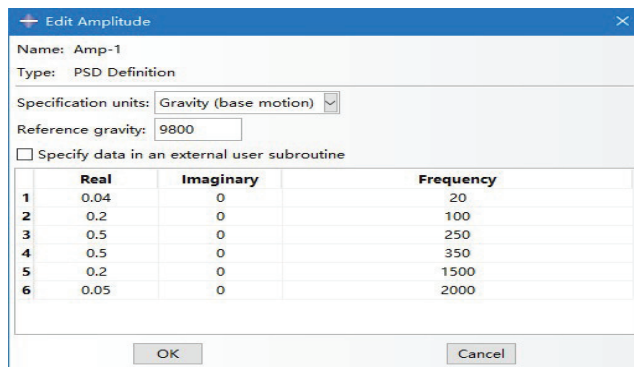


Figure 8. Establishing the input PSD curve

### B. Output result

The acceleration power spectrum of the structural vibration environment ranges from 20Hz to 2000Hz. Based on the modal analysis, the above random excitation is applied along the positive direction of the Z axis to the constraint boundary of the analysis object, that is, the fixture base. Since we marked the output of key locations in the acceleration sensor map during the test, the power spectral density analysis function can be used to obtain the power spectral density response at the specified position of the analysis object, including the acceleration power spectral density response and the total acceleration power spectral density. response. Through the calculation tool in ABAQUS, the total acceleration power spectral density integral and root mean square can be obtained [10].

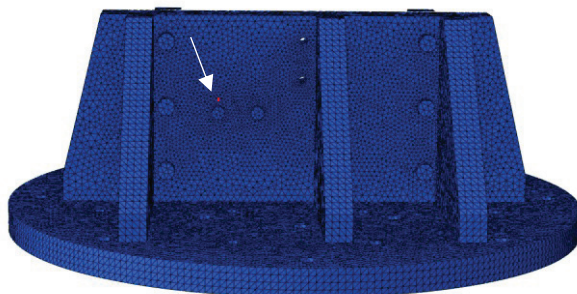


Figure 9. Location of key nodes (red dots)

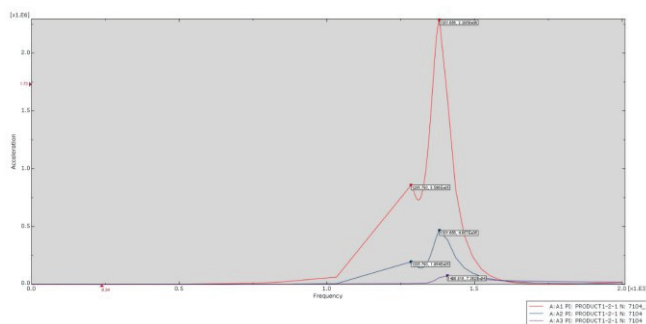


Figure 10. Acceleration response curves of key nodes in X, Y, and Z directions

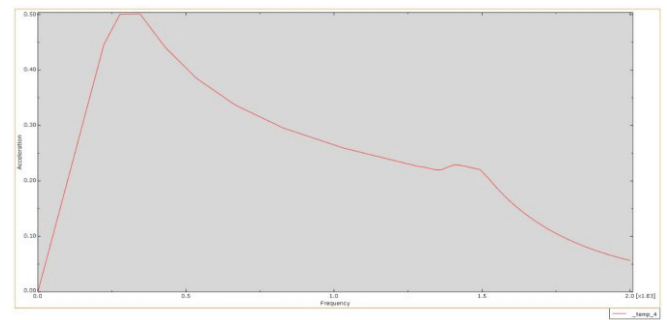


Figure 11. The total acceleration response curve in the Z direction of key nodes

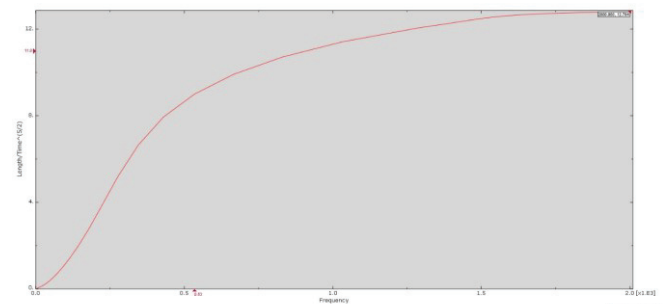


Figure 12. Root mean square value of total acceleration in Z direction of key nodes

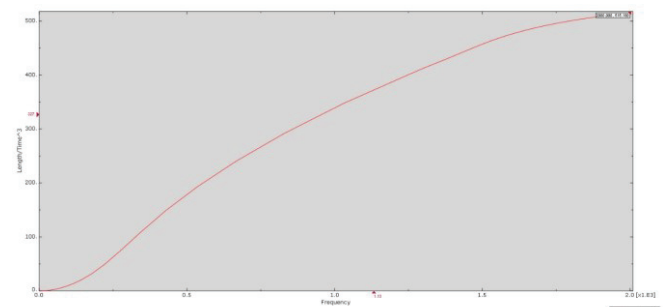


Figure 13. The integral value of the total acceleration response curve in the Z direction of the key node

### C. Analysis result

It can be seen from Fig. 10 that in the acceleration response of the key nodes, the X-axis and Y-axis directions have obvious resonance at 1285 Hz and 1381 Hz respectively, and the Z-axis direction has slight resonance at 1408 Hz. This is consistent with the results of modal analysis. Obvious resonance occurs at the first-order natural frequency and the second-order natural frequency respectively. In the real test, attention should be paid to these two frequency positions. Since the excitation application direction is the Z-axis direction, comparing the Z-axis total acceleration response curve with the input PSD curve, it can be found that the input and output stages are consistent, indicating that the fixture transfers energy well and meets the expected goal.

## V. FIXTURE BOLT LAYOUT INFLUENCE ANALYSIS

In the initial design, the intervals between the bolts are set to 50mm. Now do two sets of control simulation experiments, set the fixture bolt interval to 30mm and 80mm respectively, and conduct dynamic analysis including modal analysis and random response analysis. The results are as follows.

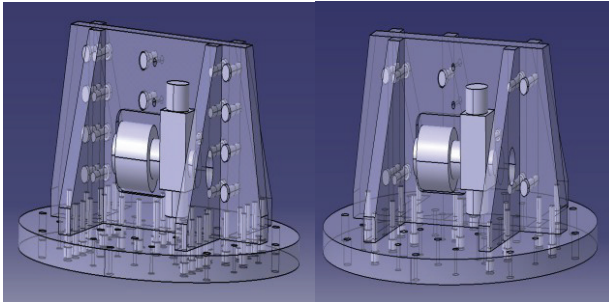


Figure 14. Structure drawing with an interval of 30mm and 80mm

### A. Modal analysis results

TABLE III. THE FIRST TEN NATURAL FREQUENCIES OF THE BOLTS WITH AN INTERVAL OF 30MM (UNIT Hz)

| Degree    | 1      | 2      | 3      | 4      | 5      |
|-----------|--------|--------|--------|--------|--------|
| Frequency | 1286.0 | 1381.8 | 2854.8 | 3197.5 | 3509.1 |
| Degree    | 6      | 7      | 8      | 9      | 10     |
| Frequency | 3611.3 | 3801.1 | 3921.3 | 4047.9 | 4132.2 |

TABLE IV. THE FIRST TEN NATURAL FREQUENCIES OF THE BOLTS WITH AN INTERVAL OF 80MM (UNIT Hz)

| Degree    | 1      | 2      | 3      | 4      | 5      |
|-----------|--------|--------|--------|--------|--------|
| Frequency | 1285.8 | 1381.6 | 2857.8 | 3215.4 | 3511.8 |
| Degree    | 6      | 7      | 8      | 9      | 10     |
| Frequency | 3612.2 | 3801.1 | 3809.6 | 3929.9 | 4050.8 |

### B. Random response analysis results

Select the same key points as in the previous random response analysis, and output the acceleration response in the X, Y, and Z directions and the total acceleration response in the Z direction with an interval of 30mm and an interval of 80mm. The difference between the results is very small. The following table only compares the curve integral and root mean square of the total additive velocity of Z-axis.

TABLE V. COMPARISON OF RESULTS

| Bolt interval | Z-axis total acceleration curve integral | Z-axis total acceleration root mean square value |
|---------------|------------------------------------------|--------------------------------------------------|
| 30            | 516.2                                    | 12.81                                            |
| 50            | 515.2                                    | 12.80                                            |
| 80            | 513.5                                    | 12.74                                            |

## C. Analysis result

From the above modal analysis and random response analysis, it can be seen that when the connection is tight, the bolt layout has little effect on the two, so we should use as few bolts as possible in the fixtures used in the test. It is convenient for processing and installation, and does not affect the transmission of external excitation to the valve.

## VI. CONCLUSION

In this paper, according to the function and use requirements of the aerospace valve, the test fixture is designed by means of T type and bolt connection. Using CATIA software, a three-dimensional solid model of fixture and a space valve structure was established. Using the theory of dynamic modal analysis and the Lanczos algorithm in the finite element software ABAQUS, the dynamic characteristics of the structure were analyzed and studied, and the modal parameters of the structure - natural frequency and mode shape were extracted. Then, based on the random vibration theory, the random response analysis of the structure is carried out, and the three-direction acceleration response curve, the vertical direction of the total additive velocity response curve, the total additive velocity root mean square and the integral of the key position are obtained. Through the analysis of simulation results, the fixture can meet the test requirements, the transmission of external excitation performance is good. By comparing the dynamic analysis results of different bolt layouts, there is little difference. Therefore, we should try to increase the bolt spacing as much as possible on the premise of ensuring tight connection.

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